Awantika Srivastava

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Professional Summary

Machine Learning Engineer with 2.5+ years of experience designing **LLM-powered application**, prompt **architectures**, and **agentic** AI systems. Strong expertise in prompt engineering, RAG pipelines, context engineering, and LLM orchestration frameworks (Langchain). Experienced in building end-to-end AI systems integrating enterprise data sources, APIs, and multi-agent workflows. Proven ability to design systems from raw data, structured outputs, automated AI-driven actions.

Professional Experience

ML Engineer *Jan 2024 – Present*
PPS International Pvt Ltd, Noida

- Designed and deployed a **real-time Driver Monitoring System (DMS)** using CNN-based object detection models (**SSD MobileNet**) to detect unsafe driving behaviors such as drowsiness, distraction, and mobile usage.
- Built **end-to-end video analytics pipelines** including frame extraction, preprocessing, model inference, and event detection for continuous monitoring.
- Developed **scalable backend services (Python, FastAPI)** to integrate ML models with monitoring dashboards and real-time alerting systems.
- Optimized deep learning models using **TensorFlow Lite, ONNX, and inference optimization techniques**, achieving **20–25 FPS with <50ms latency** on edge devices.
- Implemented **robust evaluation frameworks** using metrics such as Precision, Recall, F1-score, IoU, and mAP to ensure high model performance and reliability.
- Designed and implemented **LLM-powered applications** using **RAG pipelines, embeddings, and prompt engineering** for domain-specific question answering.
- Built **context-aware AI systems** capable of semantic retrieval, reasoning, and structured response generation.
- Developed **agentic AI workflows** using LangChain and LangGraph for multi-step reasoning, tool usage, and autonomous task execution.

Projects

Railway Driver Assistance System (RDAS)

- Built a **real-time driver monitoring system** using CNN-based SSD MobileNet to detect unsafe behaviors (drowsiness, distraction, mobile usage).
- Developed **end-to-end video pipelines** (frame extraction, preprocessing, inference, event detection) for continuous monitoring.
- Optimized models using **TensorFlow Lite & ONNX**, achieving **20–25 FPS with <50ms latency** on edge devices.
- Created a **Flask dashboard** with automated event clipping (30-sec), reducing manual review effort and improving monitoring efficiency.

LLM-Powered Domain-Specific Chatbot (RAG-based) – Railway Systems

- Developed an LLM-powered chatbot using **TinyLLaMA** to answer domain-specific queries related to railway and metro systems.
- Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS** to retrieve relevant information from technical documents such as railway regulations, PIS system specifications, and project requirement manuals.
- Processed and indexed multiple PDFs (including operational guidelines, compliance rules, and embedded system documentation) to enable accurate and context-aware responses.
- Optimized model selection (TinyLLaMA) for **low-latency inference and minimal memory usage**, suitable for deployment in resource-constrained embedded environments.

Education

Bachelor of Technology *2016 – 2020*
IMS Engineering College, Ghaziabad

Technical Skills

Languages & Frameworks: Python, C++, SQL., PySpark, Pandas, Numpy, Scikit-learn, Matplotlib, FastAPI, REST APIs

Statistics & Mathematics: EDA, Statistical Modeling, Descriptive Statistics, Hypothesis Testing, A/B Testing, Probability, Sampling, Scenario Analysis

Machine Learning: Supervised &Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).

Deep Learning: Neural Networks, CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, TensorFlow, Keras, PyTorch, TF-Lite

Time Series & Forecasting: ARIMA, SARIMA, SARIMAX, Prophet, Demand Forecasting, Trend & Seasonality Analysis, Time Series Regression

LLM & Generative AI: LLMs, Transformers (GPT, LLaMA), HuggingFace, Prompt Engineering, Fine-tuning, Context Optimization, LLM APIs, OpenAI

Agentic AI & Orchestration: LangChain, LangGraph, Agent Workflows, Execution Chains

RAG & NLP Systems: RAG, Embeddings, Semantic Search, Context Retrieval, Text Processing, Tokenization, Sentiment Analysis, NLTK, TF-IDF, Spacy, Chunking,

Computer Vision: Image Classification &Preprocessing, Image Segmentation, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV, Real-Time Processing, Paddle OCR

Cloud & MLOps: AWS (S3, EC2, SageMaker), CI/CD (GitHub Actions), Docker, MLflow, Real-Time Inference Pipelines

Tools & Collaboration: Git, GitHub, Agile, Streamlit , PowerBI, ONNX, Tensor-RT, Jupyter Notebook